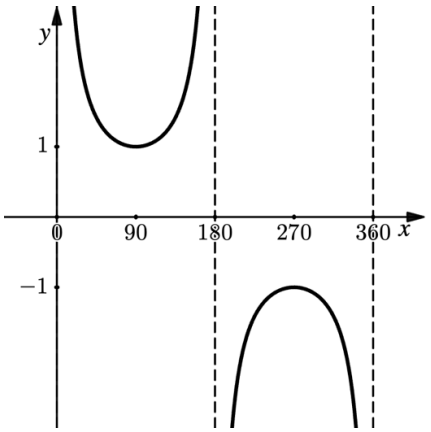
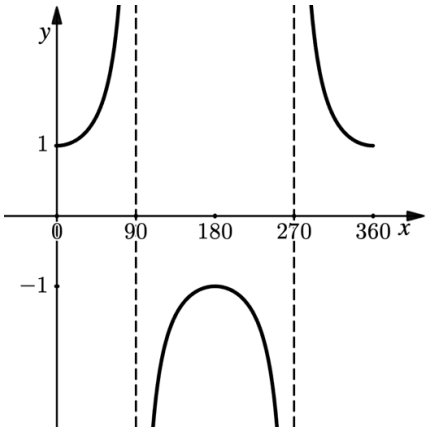
2.15 Trigonometry: Reciprocal Trigonometric Functions

Worked Solutions

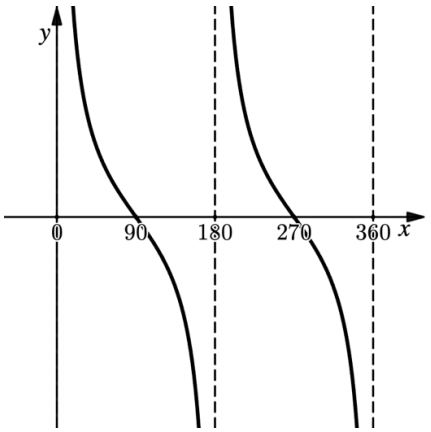
1 a)



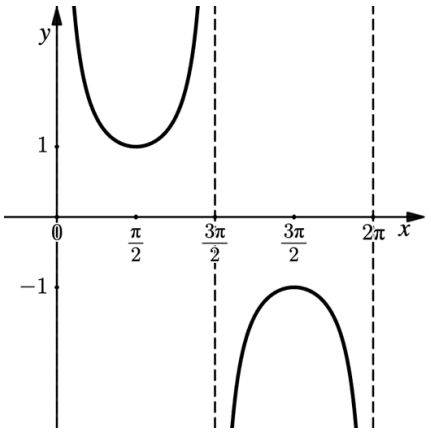
1 b)



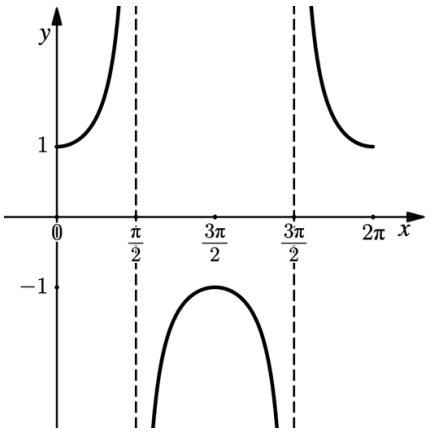
1 c)



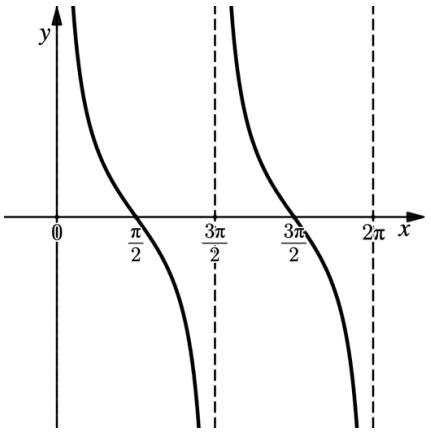
2 a)



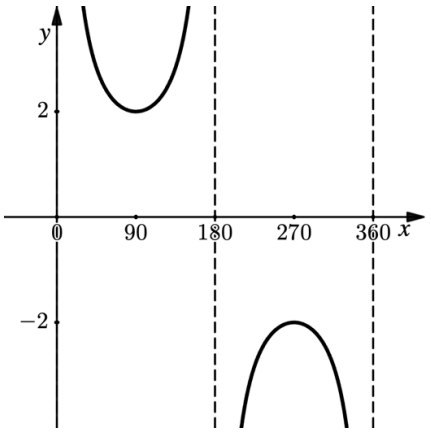
2 b)



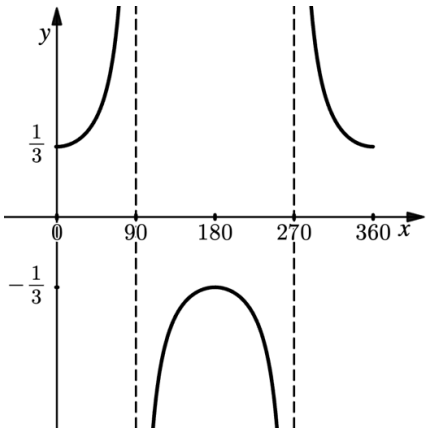
2 c)



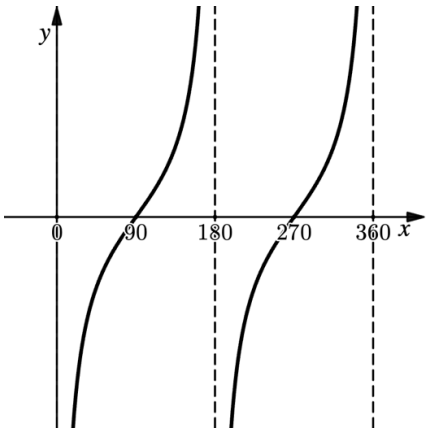
3 a)



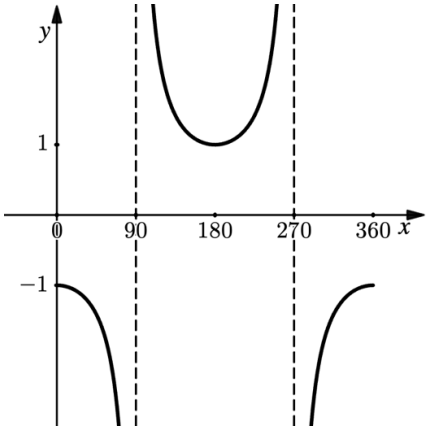
3 b)



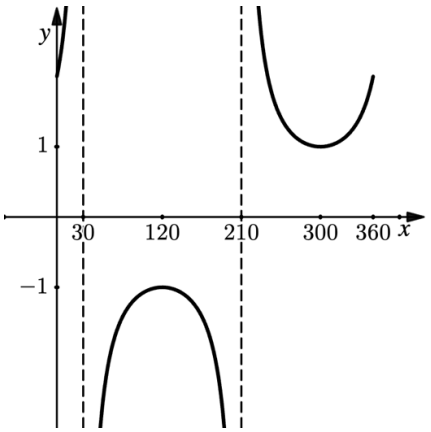
3 c)



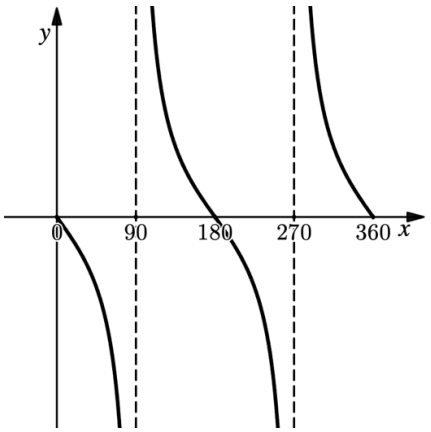
3 d)



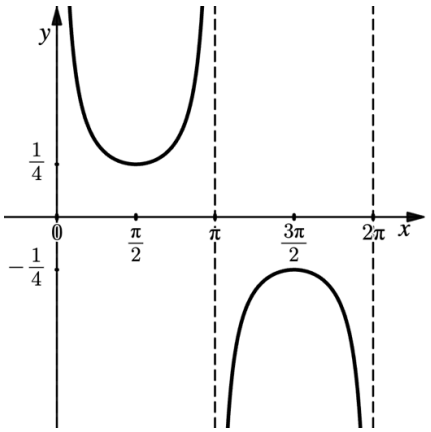
3 e)



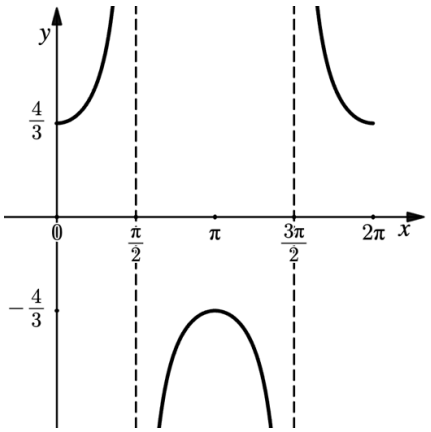
3 f)



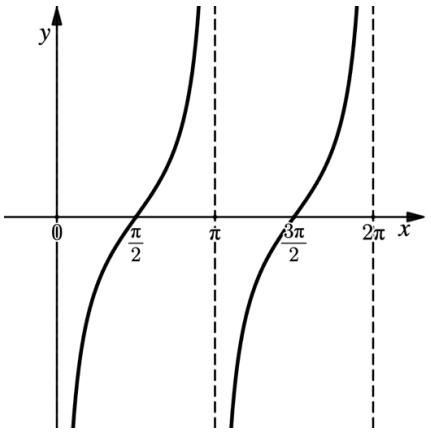
4 a)



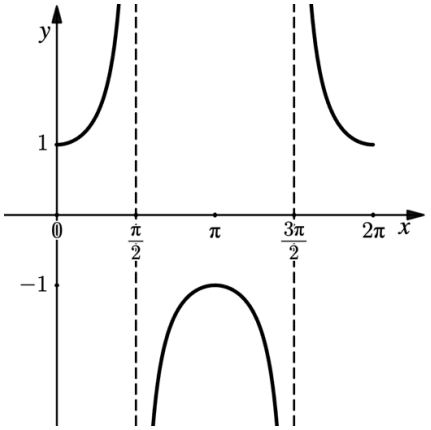
4 b)



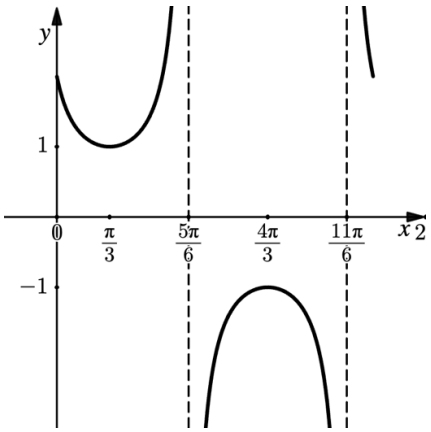
4 c)



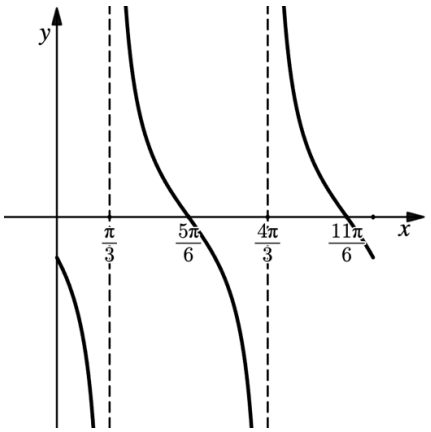
4 d)



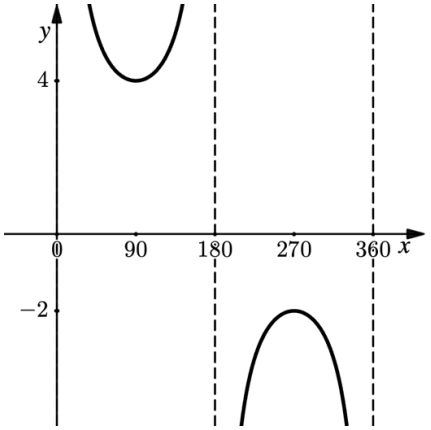
4 e)



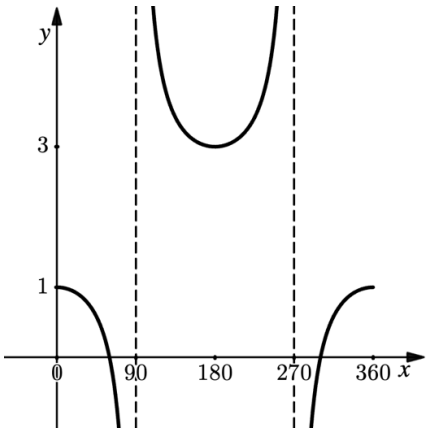
4 f)


y
x
$\frac{\pi}{3}$
$\frac{5\pi}{6}$
$\frac{4\pi}{3}$
$\frac{11\pi}{6}$

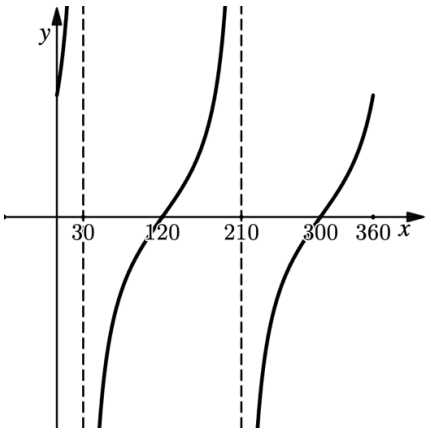
5 a)


y
4
0
90
180
270
360
x
−2

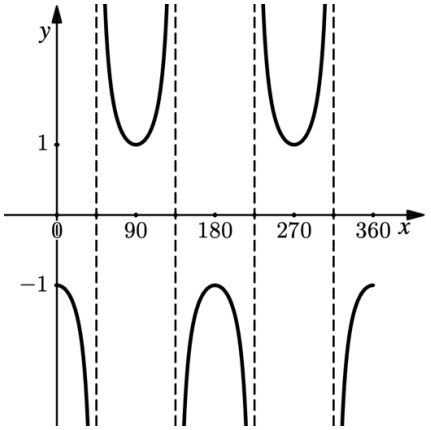
5 b)


y
3
1
0
90
180
270
360
x

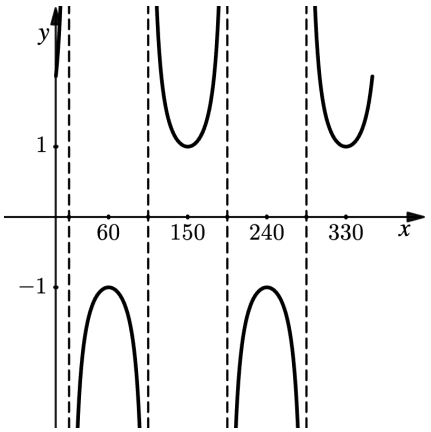
5 c)



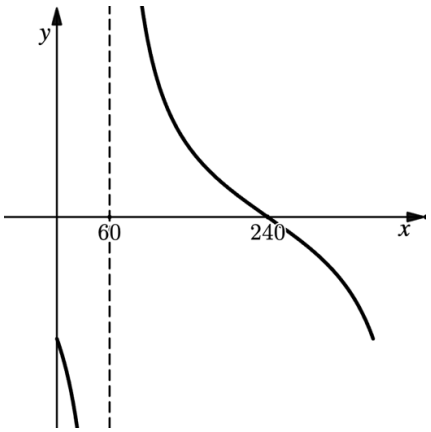
5 d)



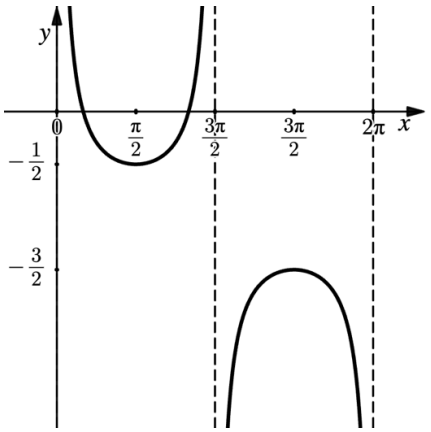
5 e)



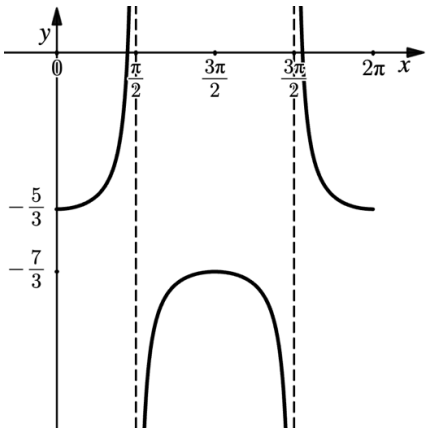
5 f)


y
60
240
x

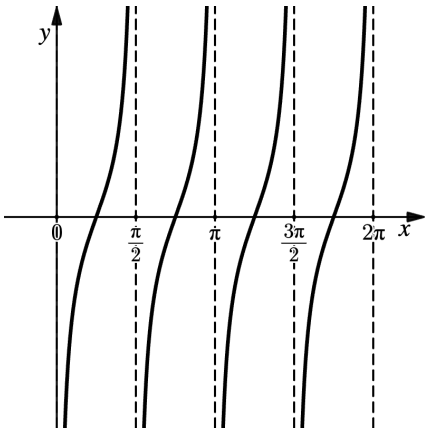
6 a)


y
0
$\frac{\pi}{2}$
$\frac{3\pi}{2}$
$\frac{3\pi}{2}$
2π
x
$-\frac{1}{2}$
$-\frac{3}{2}$

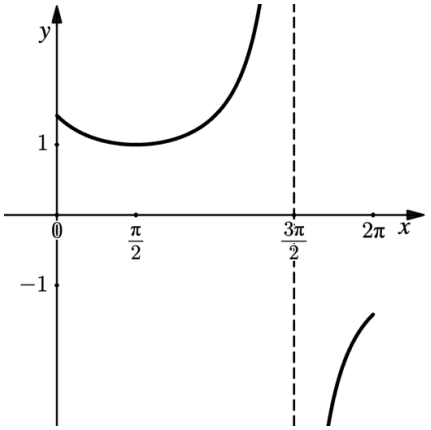
6 b)


y
0
$\frac{\pi}{2}$
$\frac{3\pi}{2}$
$\frac{3\pi}{2}$
2π
x
$-\frac{5}{3}$
$-\frac{7}{3}$

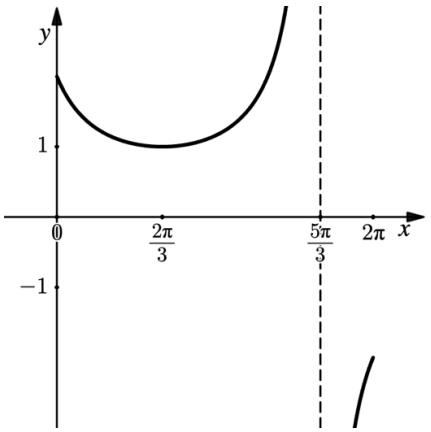
6 c)



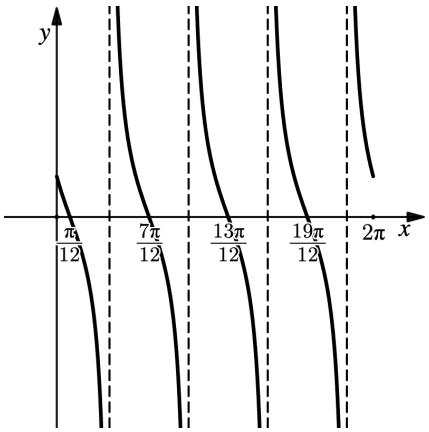
6 d)



6 e)



6 f)



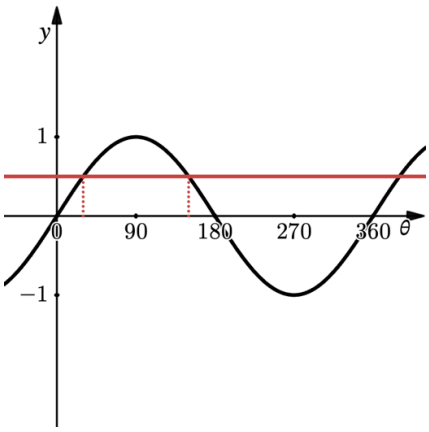
7 a)

$\operatorname{cosec}\theta = 2$ for $0° < \theta < 360°$

$\Rightarrow \frac{1}{\sin\theta} = 2 \Rightarrow \sin\theta = \frac{1}{2}$

Find first value between 0° and 360°:

$\theta = \arcsin\frac{1}{2} = 30°$ (in range)



Find second value between 0° and 360°:

$\theta = 180° - 30° = 150°$ (in range)

$\theta = \{30°, 150°\}$

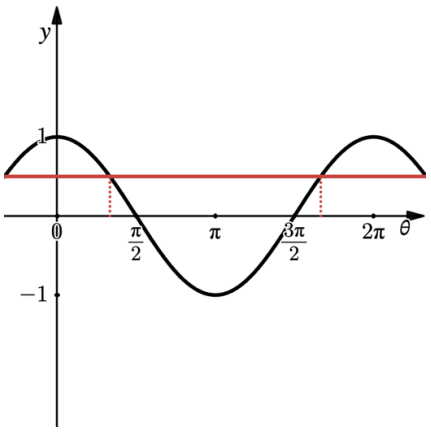
7 b)

$\sec\theta = 3$ for $0 < \theta < 2\pi$

$\Rightarrow \frac{1}{\cos\theta} = 3 \Rightarrow \cos\theta = \frac{1}{3}$

Find first value between 0 and 2π:

$\theta = \arccos\frac{1}{3} = 1.2309\ldots$ (in range)



Find second value between 0 and 2π:

$\theta = 2\pi - 1.2309\ldots = 5.0522\ldots$ (in range)

$\Rightarrow \theta = \{1.23, 5.05\}$ (3 SF)

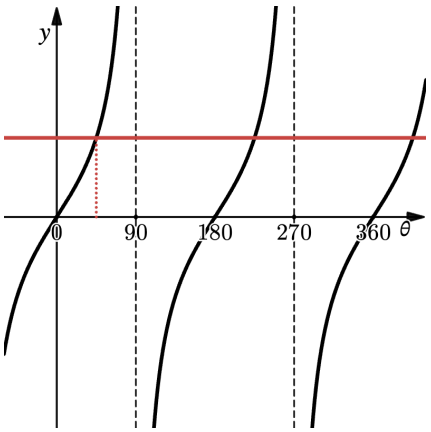
7 c)

$\cot\theta = 4$ for $0° < \theta < 360°$

$\Rightarrow \frac{1}{\tan\theta} = 4 \Rightarrow \tan\theta = \frac{1}{4}$

Find first value:

$\theta = \arctan\frac{1}{4} = 14.036\ldots°$ (in range)



Add period:

$\theta = 14.036\ldots° + 180° = 194.036\ldots°$ (in range)

$\theta = \{14.0°, 194°\}$ (3 SF)

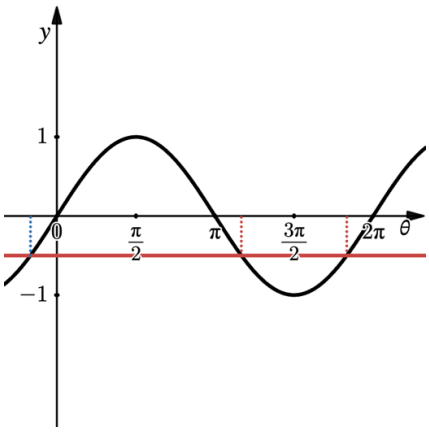
7 d)

$\operatorname{cosec}\theta = -\frac{5}{4}$ for $0 < \theta < 2\pi$

$\Rightarrow \frac{1}{\sin\theta} = -\frac{5}{4} \Rightarrow \sin\theta = -\frac{4}{5}$

Find first value:

$\theta = \arcsin\left(-\frac{4}{5}\right) = -0.9272\ldots$ (not in range)



Find first value between 0 and 2π:

$\theta = \pi + 0.9272\ldots = 4.0688\ldots$ (in range)

Find second value between 0 and 2π:

$\theta = 2\pi - 0.9272\ldots = 5.3558\ldots$ (in range)

$\theta = \{4.07, 5.36\}$ (3 SF)

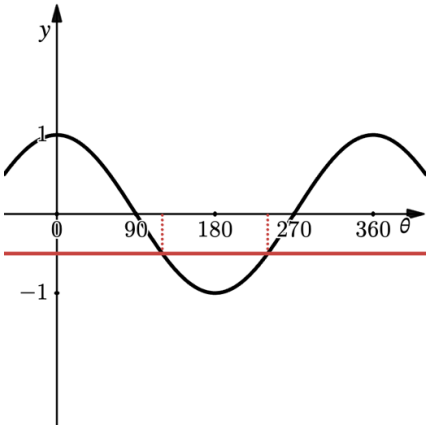
7 e)

$\sec\theta = -\frac{9}{2}$ for $0° < \theta < 360°$

$\Rightarrow \frac{1}{\cos\theta} = -\frac{9}{2} \Rightarrow \cos\theta = -\frac{2}{9}$

Find first value between 0° and 360°:

$\theta = \arccos\left(-\frac{2}{9}\right) = 102.839\ldots°$ (in range)



Find second value between 0° and 360°:

$\theta = 360° - 102.839\ldots° = 257.160\ldots°$ (in range)

$\Rightarrow \theta = \{103°, 257°\}$ (3 SF)

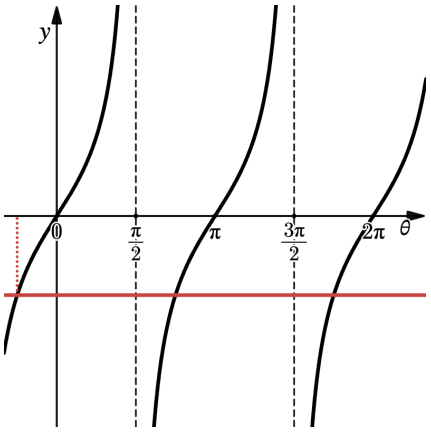
7 f)

$\cot\theta = -\frac{18}{5}$ for $0 < \theta < 2\pi$

$\Rightarrow \frac{1}{\tan\theta} = -\frac{18}{5} \Rightarrow \tan\theta = -\frac{5}{18}$

Find first value:

$\theta = \arctan\left(-\frac{5}{18}\right) = -0.2709\ldots$ (not in range)



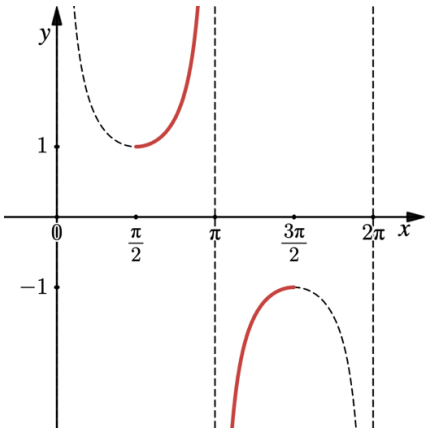
Add on period:

$\theta = -0.2709\ldots + \pi = 2.8706\ldots$ (in range)

$\theta = 2.8706\ldots + \pi = 6.0122\ldots$ (in range)

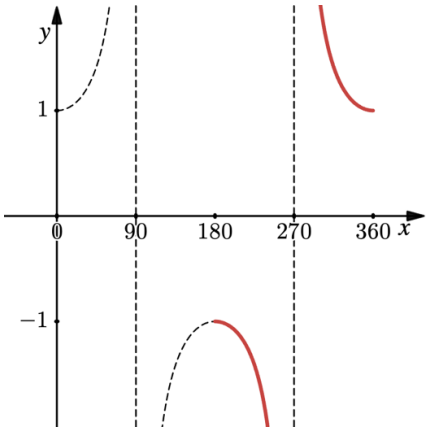
$\Rightarrow \theta = \{2.87, 6.01\}$ (3 SF)

8)



$\frac{\pi}{2} < x < \pi$ or $\pi < x < \frac{3\pi}{2}$

9)



$180° < x < 270°$ or $270° < x < 360°$

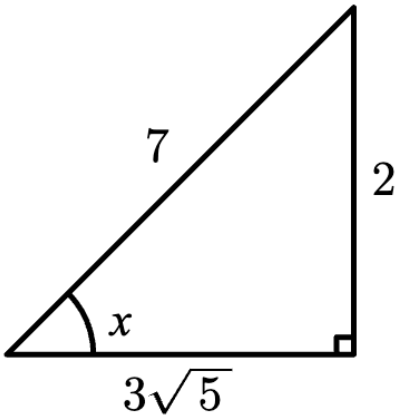
10 a)

$\operatorname{cosec} x = \frac{7}{2} \Rightarrow \frac{1}{\sin x} = \frac{7}{2} \Rightarrow \sin x = \frac{2}{7}$

10 b)

x is **ACUTE**

Draw a triangle with this information:



$\cos x = \frac{3\sqrt{5}}{7}$

10 c)

$\tan x = \frac{2}{3\sqrt{5}} = \frac{2\sqrt{5}}{15}$

10 d)

$\sec x = \frac{1}{\cos x} = \frac{1}{\frac{3\sqrt{5}}{7}} = \frac{7}{3\sqrt{5}} = \frac{7\sqrt{5}}{15}$

10 e)

$\cot x = \frac{1}{\tan x} = \frac{1}{\frac{2}{3\sqrt{5}}} = \frac{3\sqrt{5}}{2}$

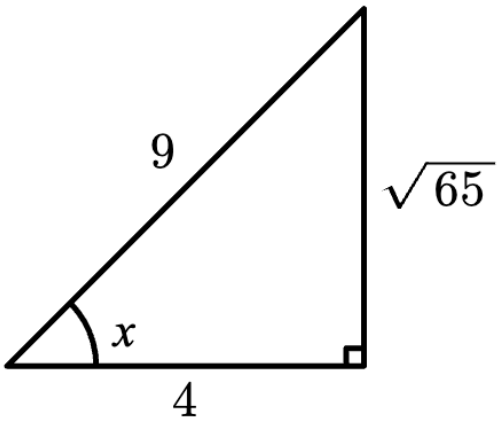
11 a)

$\sec x = \frac{9}{4} \Rightarrow \frac{1}{\cos x} = \frac{9}{4} \Rightarrow \cos x = \frac{4}{9}$

11 b)

x is **ACUTE**

Draw a triangle with this information:



$\sin x = \frac{\sqrt{65}}{9}$

11 c)

$\tan x = \frac{\sqrt{65}}{4}$

11 d)

$\text{cosec}\, x = \frac{1}{\sin x} = \frac{1}{\frac{\sqrt{65}}{9}} = \frac{9}{\sqrt{65}}$

11 e)

$\cot x = \frac{1}{\tan x} = \frac{1}{\frac{\sqrt{65}}{4}} = \frac{4}{\sqrt{65}} = \frac{4\sqrt{65}}{65}$

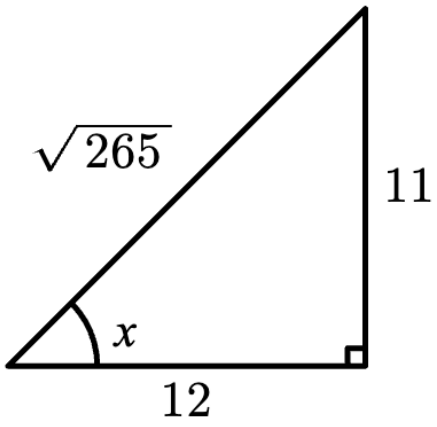
12 a)

$\cot x = \frac{12}{11} \Rightarrow \frac{1}{\tan x} = \frac{12}{11} \Rightarrow \tan x = \frac{11}{12}$

12 b)

x is **ACUTE**

Draw a triangle with this information:



$\sin x = \frac{11}{\sqrt{265}} = \frac{11\sqrt{265}}{265}$

12 c)

$\cos x = \frac{12}{\sqrt{265}} = \frac{12\sqrt{265}}{265}$

12 d)

$\text{cosec}\, x = \frac{1}{\sin x} = \frac{1}{\frac{11}{\sqrt{265}}} = \frac{\sqrt{265}}{11}$

12 e)

$\sec x = \frac{1}{\cos x} = \frac{1}{\frac{12}{\sqrt{265}}} = \frac{\sqrt{265}}{12}$

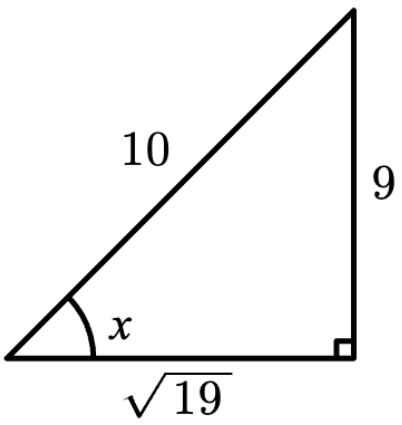
13 a)

$\text{cosec}\, x = \frac{10}{9} \Rightarrow \frac{1}{\sin x} = \frac{10}{9} \Rightarrow \sin x = \frac{9}{10}$

13 b)

x is **OBTUSE**

Ignore $90° < x < 180°$ for the moment and draw a right-angled triangle with this information:



$\cos x = -\frac{\sqrt{19}}{10}$ (the result is negative as $\cos x$ is below the x-axis between $90°$ and $180°$)

13 c)

$\tan x = -\frac{9}{\sqrt{19}} = -\frac{9\sqrt{19}}{19}$ (the result is negative as $\tan x$ is below the x-axis between 90° and 180°)

13 d)

$\sec x = \frac{1}{\cos x} = \frac{1}{-\frac{\sqrt{19}}{10}} = -\frac{10}{\sqrt{19}} = -\frac{10\sqrt{19}}{19}$

13 e)

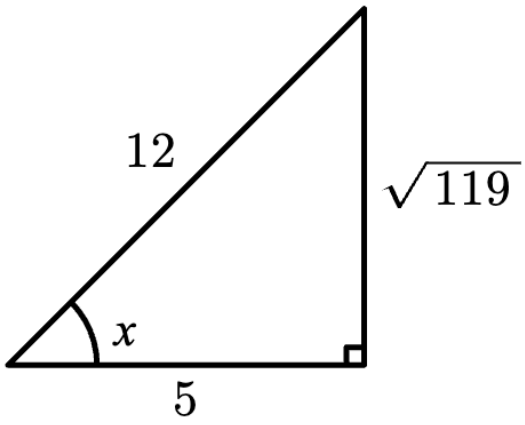
$\cot x = \frac{1}{\tan x} = \frac{1}{-\frac{9}{\sqrt{19}}} = -\frac{\sqrt{19}}{9}$

14 a)

$\sec x = -\frac{12}{5} \Rightarrow \frac{1}{\cos x} = -\frac{12}{5} \Rightarrow \cos x = -\frac{5}{12}$

14 b)

Ignore $180° < x < 270°$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\sin x = -\frac{\sqrt{119}}{12}$ (the result is negative as $\sin x$ is below the x-axis between 180° and 270°)

14 c)

$\tan x = \frac{\sqrt{119}}{5}$ (the result is positive as $\tan x$ is above the x-axis between 180° and 270°)

14 d)

$\operatorname{cosec} x = \frac{1}{\sin x} = \frac{1}{-\frac{\sqrt{119}}{12}} = -\frac{12}{\sqrt{119}} = -\frac{12\sqrt{119}}{119}$

14 e)

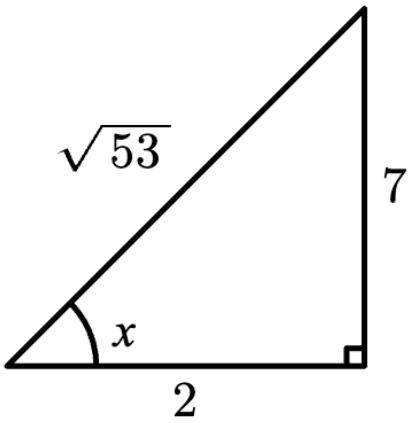
$\cot x = \frac{1}{\tan x} = \frac{1}{\frac{\sqrt{119}}{5}} = \frac{5}{\sqrt{119}} = \frac{5\sqrt{119}}{119}$

15 a)

$\cot x = -\frac{2}{7} \Rightarrow \frac{1}{\tan x} = -\frac{2}{7} \Rightarrow \tan x = -\frac{7}{2}$

15 b)

Ignore $270° < x < 360°$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\sin x = -\frac{7}{\sqrt{53}} = -\frac{7\sqrt{53}}{53}$ (the result is negative as $\sin x$ is below the x-axis between 270° and 360°)

15 c)

$\cos x = \frac{2}{\sqrt{53}} = \frac{2\sqrt{53}}{53}$ (the result is positive as $\cos x$ is above the x-axis between 270° and 360°)

15 d)

$\operatorname{cosec} x = \frac{1}{\sin x} = \frac{1}{-\frac{7}{\sqrt{53}}} = -\frac{\sqrt{53}}{7}$

15 e)

$\sec x = \frac{1}{\cos x} = \frac{1}{\frac{2}{\sqrt{53}}} = \frac{\sqrt{53}}{2}$

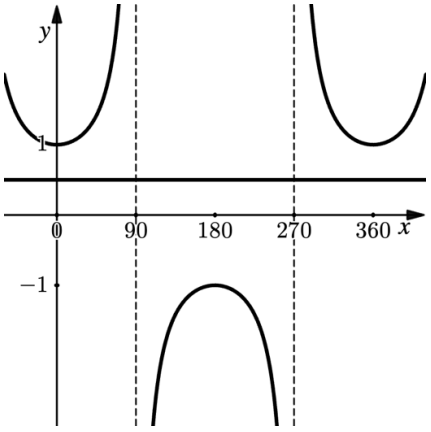
16 a)

$8\sec^2 x - 14\sec x + 3 = 0$

$\Rightarrow (4\sec x - 1)(2\sec x - 3) = 0$

$\Rightarrow \sec x = \frac{1}{4}$ or $\sec x = \frac{3}{2}$

$\sec x = \frac{1}{4}$ has no solutions



So $\sec x = \frac{3}{2}$ only

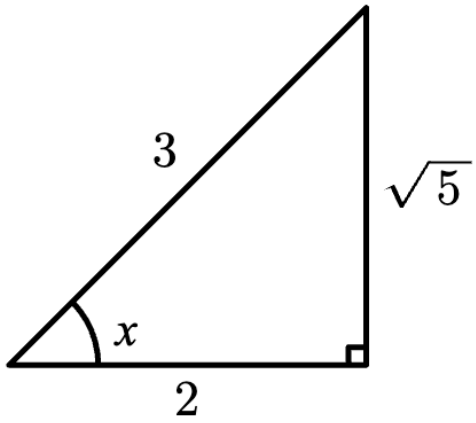
16 b)

$\sec x = \frac{3}{2} \Rightarrow \frac{1}{\cos x} = \frac{3}{2} \Rightarrow \cos x = \frac{2}{3}$

16 c)

x is **ACUTE**

Draw a triangle with this information:



$\sin x = \frac{\sqrt{5}}{3}$

16 d)

$\tan x = \frac{\sqrt{5}}{2}$

16 e)

$\operatorname{cosec} x = \frac{1}{\sin x} = \frac{1}{\frac{\sqrt{5}}{3}} = \frac{3}{\sqrt{5}} = \frac{3\sqrt{5}}{5}$

16 f)

$\cot x = \frac{1}{\tan x} = \frac{1}{\frac{\sqrt{5}}{2}} = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$

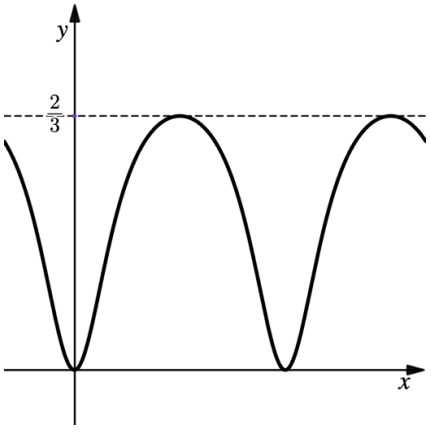
17)

$y = \operatorname{cosec} x$ only exists for $y \leq -1$ or $y \geq 1$

So $y = \operatorname{cosec}^2 x$ only exists for $y \geq 1$

So the minimum value of $\operatorname{cosec}^2 x$ is 1

So the maximum value of $\frac{2}{\operatorname{cosec}^2 x + 2}$ is $\frac{2}{1+2} = \frac{2}{3}$



18 a)

$\frac{\sin x}{\cos^2 x} \equiv \frac{\sin x}{\cos x} \times \frac{1}{\cos x} \equiv \tan x \sec x$

18 b)

$\frac{\cos x}{\sin^2 x} \equiv \frac{\cos x}{\sin x} \times \frac{1}{\sin x} \equiv \cot x \operatorname{cosec} x$

19 a)

$\operatorname{cosec} x \sin x \equiv \frac{1}{\sin x} \times \sin x \equiv 1$

19 b)

$\sec x \cos x \equiv \frac{1}{\cos x} \times \cos x \equiv 1$

19 c)

$\cot x \tan x \equiv \frac{1}{\tan x} \times \tan x \equiv 1$

20 a)

$\operatorname{cosec} x \sec x \equiv \frac{1}{\sin x} \times \frac{1}{\cos x} \equiv \frac{1}{\sin x \cos x}$

20 b)

$\operatorname{cosec} x \cot x \equiv \frac{1}{\sin x} \times \frac{\cos x}{\sin x} \equiv \frac{\cos x}{\sin^2 x}$

20 c)

$\sec x \cot x \equiv \frac{1}{\cos x} \times \frac{\cos x}{\sin x} \equiv \frac{1}{\sin x}$

20 d)

$\operatorname{cosec} x \sec x \cot x \equiv \frac{1}{\sin x} \times \frac{1}{\cos x} \times \frac{\cos x}{\sin x} \equiv \frac{1}{\sin^2 x}$

21 a)

$\frac{\operatorname{cosec} x}{\sec x} \equiv \frac{\frac{1}{\sin x}}{\frac{1}{\cos x}} \equiv \frac{1}{\sin x} \times \frac{\cos x}{1} = \frac{\cos x}{\sin x} = \cot x$

21 b)

$\frac{\operatorname{cosec} x}{\cot x} \equiv \frac{\frac{1}{\sin x}}{\frac{\cos x}{\sin x}} \equiv \frac{1}{\sin x} \times \frac{\sin x}{\cos x} \equiv \frac{1}{\cos x} \equiv \sec x$

21 c)

$\frac{\sec x}{\operatorname{cosec} x} \equiv \frac{\frac{1}{\cos x}}{\frac{1}{\sin x}} \equiv \frac{1}{\cos x} \times \frac{\sin x}{1} \equiv \frac{\sin x}{\cos x} \equiv \tan x$

21 d)

$\frac{\sec x}{\cot x} \equiv \frac{\frac{1}{\cos x}}{\frac{\cos x}{\sin x}} \equiv \frac{1}{\cos x} \times \frac{\sin x}{\cos x} \equiv \sec x \tan x$

21 e)

$\frac{\cot^2 x}{\sec x} \equiv \frac{\frac{\cos^2 x}{\sin^2 x}}{\frac{1}{\cos x}} \equiv \frac{\cos^2 x}{\sin^2 x} \times \frac{\cos x}{1} \equiv \cot^2 x \cos x$

21 f)

$\frac{\cot^2 x \sin^3 x}{\operatorname{cosec} x \sec x} \equiv \frac{\frac{\cos^2 x}{\sin^2 x} \times \sin^3 x}{\frac{1}{\sin x} \times \frac{1}{\cos x}} \equiv \frac{\cos^2 x \sin x}{\frac{1}{\sin x \cos x}}$

$\equiv \cos^2 x \sin x \times \frac{\sin x \cos x}{1} \equiv \cos^3 x \sin^2 x$

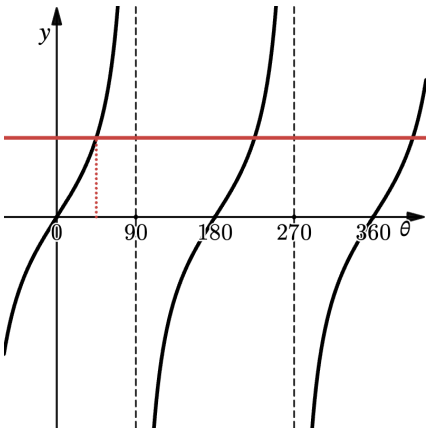
22 a)

$\operatorname{cosec} \theta = \sec \theta$ for $0° < \theta < 360°$

$\Rightarrow \frac{1}{\sin \theta} = \frac{1}{\cos \theta} \Rightarrow 1 = \frac{\sin \theta}{\cos \theta} \Rightarrow \tan \theta = 1$

Find first value:

$\theta = \arctan 1 = 45°$ (in range)



Add period:

$\theta = 45° + 180° = 225°$ (in range)

$\Rightarrow \theta = \{45°, 225°\}$

22 b)

$\cot \theta = -\sec \theta$ for $0 < \theta < 2\pi$

$\Rightarrow \frac{\cos \theta}{\sin \theta} = -\frac{1}{\cos \theta} \Rightarrow \cos^2 \theta = -\sin \theta$

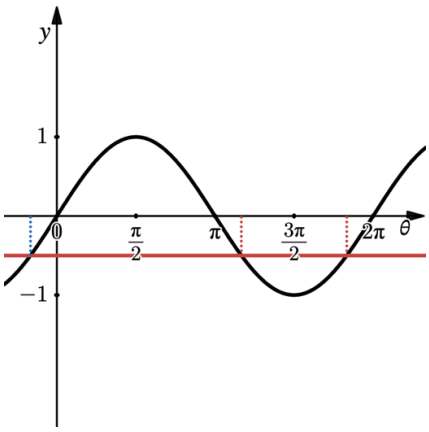
$\Rightarrow 1 - \sin^2 \theta = -\sin \theta \Rightarrow \sin^2 \theta - \sin \theta - 1 = 0$

$\Rightarrow \sin \theta = \frac{1 \pm \sqrt{5}}{2}$

First solve $\sin \theta = \frac{1-\sqrt{5}}{2}$:

Find first value:

$\theta = \arcsin\left(\frac{1-\sqrt{5}}{2}\right) = -0.6662\ldots$ (not in range)



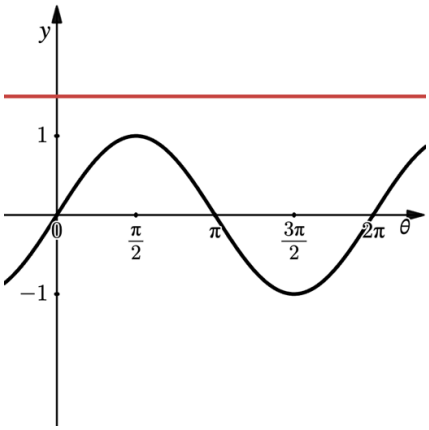
Find first value between 0 and 2π:

$\theta = \pi + 0.6662\ldots = 3.8078\ldots$ (in range)

Find second value between 0 and 2π:

$\theta = 2\pi - 0.6662\ldots = 5.6169\ldots$ (in range)

As for $\sin\theta = \frac{1+\sqrt{5}}{2}$, there are no solutions:



$\theta = \{3.81, 5.62\}$ (3 SF)

22 c)

$\sec\theta = \cot\theta$ for $0° < \theta < 360°$

$\Rightarrow \frac{1}{\cos\theta} = \frac{\cos\theta}{\sin\theta} \Rightarrow \sin\theta = \cos^2\theta$

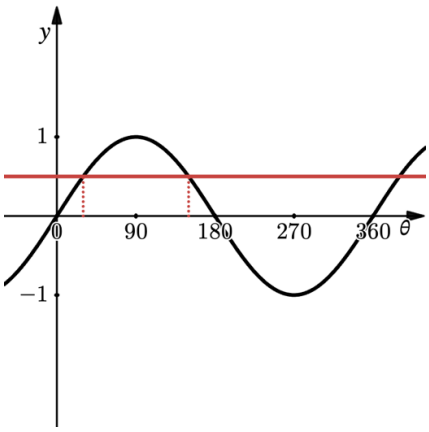
$\Rightarrow \sin\theta = 1 - \sin^2\theta \Rightarrow \sin^2\theta + \sin\theta - 1 = 0$

$\Rightarrow \sin\theta = \frac{-1\pm\sqrt{5}}{2}$

Solve $\sin\theta = \frac{-1+\sqrt{5}}{2}$ first:

Find first value between 0° and 360°:

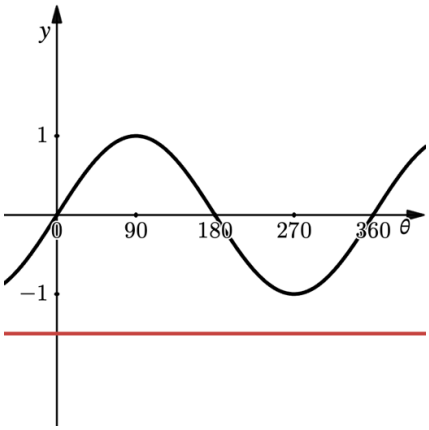
$\theta = \arcsin\left(\frac{-1+\sqrt{5}}{2}\right) = 38.172\ldots°$ (in range)



Find second value between 0° and 360°:

$\theta = 180° - 38.172\ldots° = 141.827\ldots°$ (in range)

As for $\sin\theta = \frac{-1+\sqrt{5}}{2}$, there are no solutions:



$\theta = \{38.2°, 142°\}$ (3 SF)

22 d)

$\cot\theta = \tan\theta$ for $0 < \theta < 2\pi$

$\Rightarrow \frac{\cos\theta}{\sin\theta} = \frac{\sin\theta}{\cos\theta} \Rightarrow \cos^2\theta = \sin^2\theta$

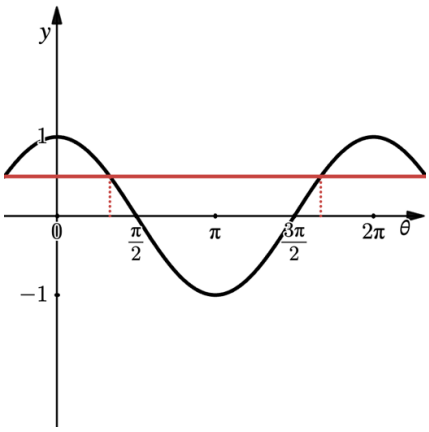
$\Rightarrow \cos^2\theta = 1 - \cos^2\theta \Rightarrow 2\cos^2\theta = 1$

$\Rightarrow \cos^2\theta = \frac{1}{2} \Rightarrow \cos\theta = \pm\frac{1}{\sqrt{2}}$

Solve $\cos\theta = \frac{1}{\sqrt{2}}$ first:

Find first value between 0 and 2π:

$\theta = \arccos\frac{1}{\sqrt{2}} = \frac{\pi}{4}$ (in range)



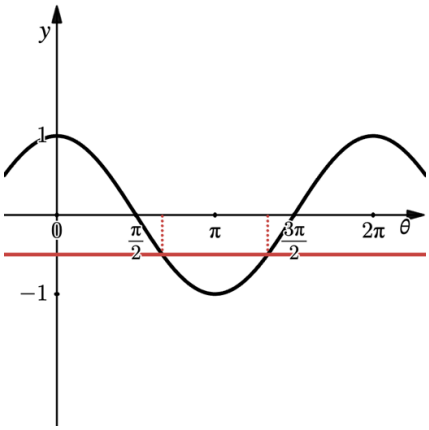
Find second value between 0 and 2π:

$\theta = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$ (in range)

Now solve $\cos\theta = -\frac{1}{\sqrt{2}}$:

Find first value between 0 and 2π:

$\theta = \arccos\left(-\frac{1}{\sqrt{2}}\right) = \frac{3\pi}{4}$ (in range)

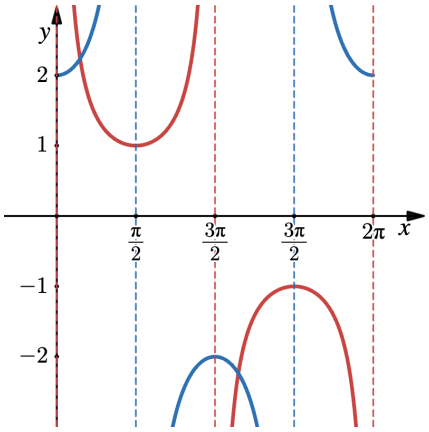


Find second value between 0 and 2π:

$\theta = 2\pi - \frac{3\pi}{4} = \frac{5\pi}{8}$ (in range)

$\Rightarrow \theta = \left\{\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}\right\}$

23 a)



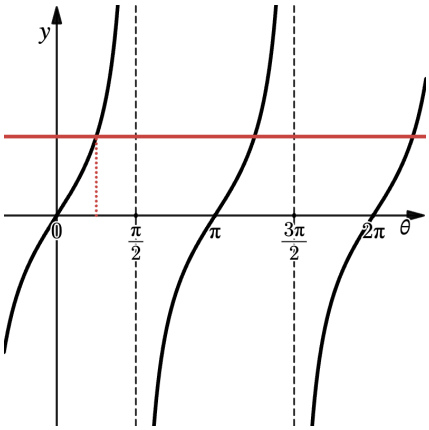
23 b)

$\operatorname{cosec} x = 2\sec x$ for $0 < x < 2\pi$

$\Rightarrow \frac{1}{\sin x} = \frac{2}{\cos x} \Rightarrow \frac{1}{2} = \frac{\sin x}{\cos x} \Rightarrow \tan x = \frac{1}{2}$

Find first value:

$\theta = \arctan \frac{1}{2} = 0.4636\ldots$ (in range)



Add period:

$\theta = 0.4636\ldots + \pi = 3.6052\ldots$ (in range)

When $\theta = 0.4636\ldots$, $y = \operatorname{cosec}(0.4636\ldots)$

$= \frac{1}{\sin(0.4636\ldots)} = 2.2360\ldots$

When $\theta = 3.6052\ldots$, $y = \operatorname{cosec}(3.6052\ldots)$

$= \frac{1}{\sin(3.6052\ldots)} = -2.2360\ldots$

So the two curves intersect at $(0.464, 2.24)$ and $(3.61, -2.24)$ (3 SF)

24)

$\sec x - 3 + k = 0 \Rightarrow \sec x = 3 - k$

$y = \sec x$ only exists for $y \leq -1$ or $y \geq 1$

So $3 - k \leq -1 \Rightarrow k \geq 4$

Or $3 - k \geq 1 \Rightarrow k \leq 2$

25 a)

$\sin^2 x + \cos^2 x \equiv 1 \Rightarrow \frac{\sin^2 x}{\sin^2 x} + \frac{\cos^2 x}{\sin^2 x} \equiv \frac{1}{\sin^2 x}$

$\Rightarrow 1 + \cot^2 x \equiv \operatorname{cosec}^2 x$

25 b)

$\sin^2 x + \cos^2 x \equiv 1 \Rightarrow \frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} \equiv \frac{1}{\cos^2 x}$

$\Rightarrow \tan^2 x + 1 \equiv \sec^2 x$

26 a)

$\frac{1+\cot^2 x}{\operatorname{cosec}^2 x} \equiv \frac{\operatorname{cosec}^2 x}{\operatorname{cosec}^2 x} \equiv 1$

26 b)

$\frac{1+\tan^2 x}{\sec^2 x} \equiv \frac{\sec^2 x}{\sec^2 x} \equiv 1$

26 c)

$\frac{1+\tan^2 x}{1+\cot^2 x} \equiv \frac{\sec^2 x}{\operatorname{cosec}^2 x} \equiv \frac{\frac{1}{\cos^2 x}}{\frac{1}{\sin^2 x}} \equiv \frac{1}{\cos^2 x} \times \frac{\sin^2 x}{1}$

$\equiv \frac{\sin^2 x}{\cos^2 x} \equiv \tan^2 x$

26 d)

$\frac{2+2\cot^2 x}{5+5\tan^2 x} \equiv \frac{2(1+\cot^2 x)}{5(1+\tan^2 x)} \equiv \frac{2\operatorname{cosec}^2 x}{5\sec^2 x} \equiv \frac{2}{5} \times \frac{\frac{1}{\sin^2 x}}{\frac{1}{\cos^2 x}}$

$\equiv \frac{2}{5} \times \frac{1}{\sin^2 x} \times \frac{\cos^2 x}{1} \equiv \frac{2}{5} \times \frac{\cos^2 x}{\sin^2 x} \equiv \frac{2}{5}\cot^2 x$

27 a)

$\cot^2\theta + 3\operatorname{cosec}\theta - 3 = 0$ for $0° < \theta < 360°$

$\Rightarrow (\operatorname{cosec}^2\theta - 1) + 3\operatorname{cosec}\theta - 3 = 0$

$\Rightarrow \operatorname{cosec}^2\theta + 3\operatorname{cosec}\theta - 4 = 0$

$\Rightarrow (\operatorname{cosec}\theta + 4)(\operatorname{cosec}\theta - 1) = 0$

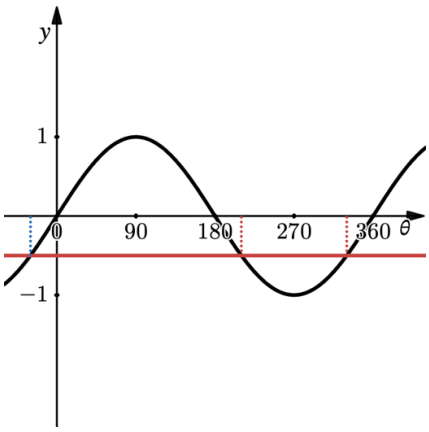
$\Rightarrow \operatorname{cosec}\theta = -4$ or $\operatorname{cosec}\theta = 1$

Solve $\operatorname{cosec}\theta = -4$ first:

$\Rightarrow \frac{1}{\sin\theta} = -4 \Rightarrow \sin\theta = -\frac{1}{4}$

Find first value:

$\theta = \arcsin\left(-\frac{1}{4}\right) = -14.477\ldots°$ (not in range)



Find first value between 0° and 360°:

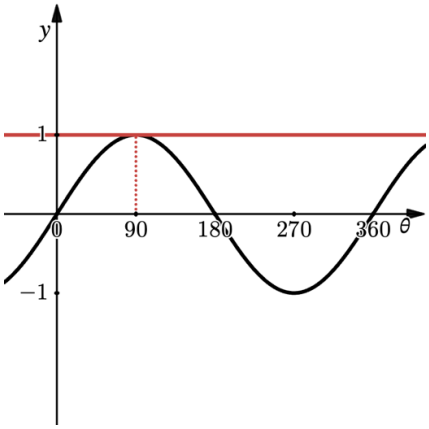
$\theta = 180° + 14.477\ldots° = 194.477\ldots°$ (in range)

Find second value between 0° and 360°:

$\theta = 360° - 14.477\ldots° = 345.522\ldots°$ (in range)

Now solve $\operatorname{cosec}\theta = 1$:

$\Rightarrow \frac{1}{\sin\theta} = 1 \Rightarrow \sin\theta = 1$



$\theta = 90°$ (in range)

$\theta = \{90°, 194°, 346°\}$ (3 SF)

27 b)

$\tan^2\theta - \sec\theta - 11 = 0$ for $0 < \theta < 2\pi$

$\Rightarrow (\sec^2\theta - 1) - \sec\theta - 11 = 0$

$\Rightarrow \sec^2\theta - \sec\theta - 12 = 0$

$\Rightarrow (\sec\theta - 4)(\sec\theta + 3) = 0$

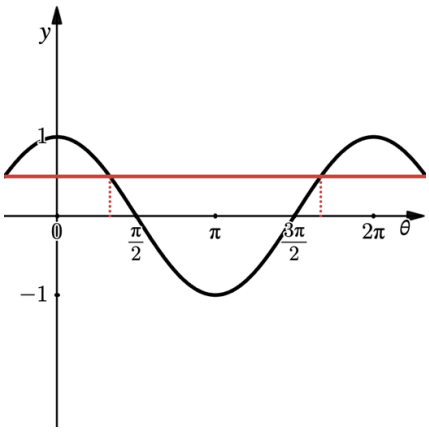
$\Rightarrow \sec\theta = 4$ or $\sec\theta = -3$

Solve $\sec\theta = 4$ first:

$\Rightarrow \frac{1}{\cos\theta} = 4 \Rightarrow \cos\theta = \frac{1}{4}$

Find first value between 0 and 2π:

$\theta = \arccos\frac{1}{4} = 1.3181\ldots$ (in range)



Find second value between 0 and 2π:

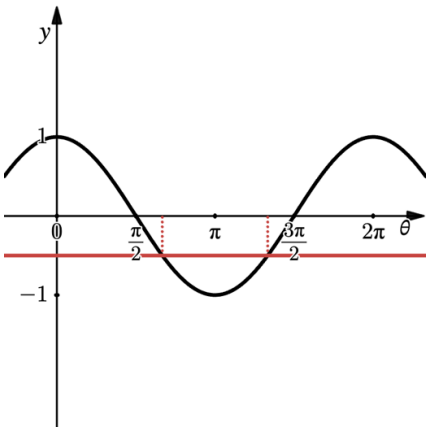
$\theta = 2\pi - 1.3181\ldots = 4.9650\ldots$ (in range)

Now solve $\sec\theta = -3$:

$\Rightarrow \frac{1}{\cos\theta} = -3 \Rightarrow \cos\theta = -\frac{1}{3}$

Find first value between 0 and 2π:

$\theta = \arccos\left(-\frac{1}{3}\right) = 1.9106\ldots$ (in range)



Find second value between 0 and 2π:

$\theta = 2\pi - 1.9106\ldots = 4.3725\ldots$ (in range)

$\Rightarrow \theta = \{1.32, 1.91, 4.37, 4.97\}$ (3 SF)

27 c)

$\operatorname{cosec}^2\theta + 5\cot\theta - 15 = 0$ for $0° < \theta < 360°$

$\Rightarrow (1 + \cot^2\theta) + 5\cot\theta - 15 = 0$

$\Rightarrow \cot^2\theta + 5\cot\theta - 14 = 0$

$\Rightarrow (\cot\theta - 2)(\cot\theta + 7) = 0$

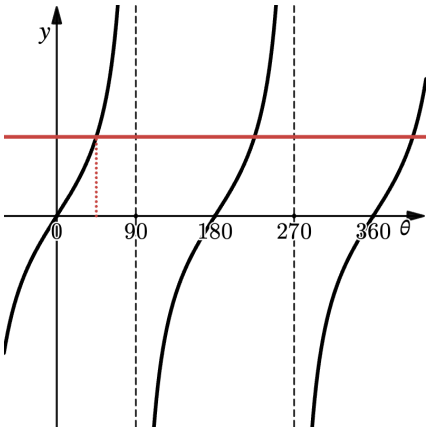
$\Rightarrow \cot\theta = 2$ or $\cot\theta = -7$

Solve $\cot\theta = 2$ first:

$\Rightarrow \frac{1}{\tan\theta} = 2 \Rightarrow \tan\theta = \frac{1}{2}$

Find first value:

$\theta = \arctan\frac{1}{2} = 26.565\ldots°$ (in range)



Add period:

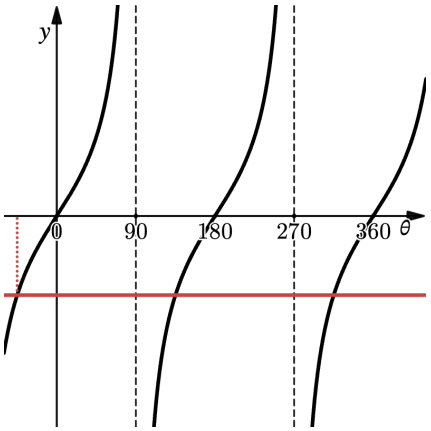
$\theta = 26.565\ldots^\circ + 180^\circ = 206.565\ldots^\circ$ (in range)

Now solve $\cot\theta = -7$:

$\Rightarrow \frac{1}{\tan\theta} = -7 \Rightarrow \tan\theta = -\frac{1}{7}$

Find first value:

$\theta = \arctan\left(-\frac{1}{7}\right) = -8.130\ldots^\circ$ (not in range)



Add period:

$\theta = -8.130\ldots^\circ + 180^\circ = 171.869\ldots^\circ$ (in range)

$\theta = 171.869\ldots^\circ + 180^\circ = 351.869\ldots^\circ$ (in range)

$\theta = \{26.6^\circ, 172^\circ, 207^\circ, 352^\circ\}$ (3 SF)

27 d)

$\sec^2\theta + 6\tan\theta - 28 = 0$ for $0 < \theta < 2\pi$

$\Rightarrow (\tan^2\theta + 1) + 6\tan\theta - 28 = 0$

$\Rightarrow \tan^2\theta + 6\tan\theta - 27 = 0$

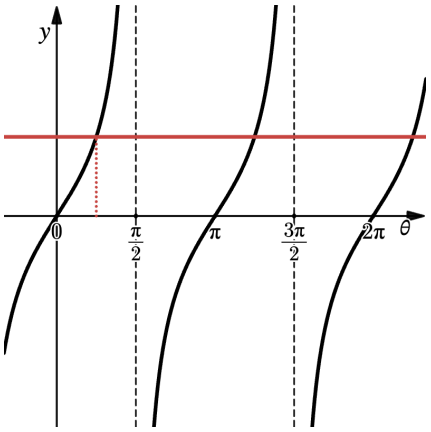
$\Rightarrow (\tan\theta - 3)(\tan\theta + 9) = 0$

$\Rightarrow \tan\theta = 3$ or $\tan\theta = -9$

Solve $\tan\theta = 3$ first:

Find first value:

$\theta = \arctan 3 = 1.2490\ldots$ (in range)



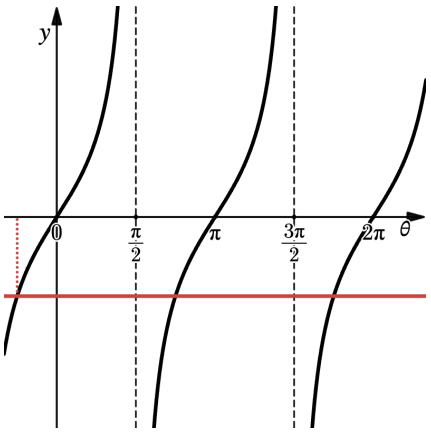
Add period:

$\theta = 1.2490\ldots + \pi = 4.3906\ldots$ (in range)

Now solve $\tan\theta = -9$:

Find first value:

$\theta = \arctan(-9) = -1.4601\ldots^\circ$ (not in range)



Add period:

$\theta = -1.4601 + \pi = 1.6814\ldots$ (in range)

$\theta = 1.6814\ldots + \pi = 4.8230\ldots$ (in range)

$\theta = \{1.25, 1.68, 4.39, 4.82\}$ (3 SF)

28 a)

$3\,\text{cosec}^2\,\theta + 2\cot\theta - 4 = 0$ for $0^\circ < \theta < 360^\circ$

$\Rightarrow 3(\cot^2\theta + 1) + 2\cot\theta - 4 = 0$

$\Rightarrow 3\cot^2\theta + 2\cot\theta - 1 = 0$

$\Rightarrow (3\cot\theta - 1)(\cot\theta + 1) = 0$

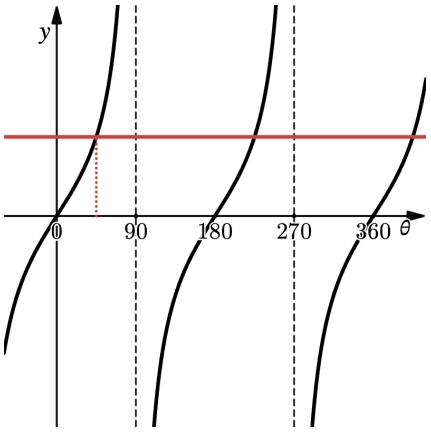
$\Rightarrow \cot\theta = \frac{1}{3}$ or $\cot\theta = -1$

Solve $\cot\theta = \frac{1}{3}$ first:

$\Rightarrow \frac{1}{\tan\theta} = \frac{1}{3} \Rightarrow \tan\theta = 3$

Find first value:

$\theta = \arctan 3 = 71.565\ldots^\circ$ (in range)



Add period:

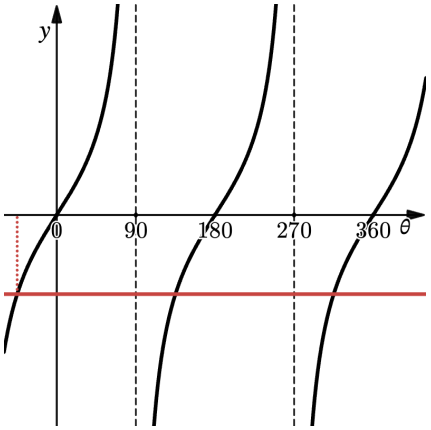
$\theta = 71.565\ldots^\circ + 180^\circ = 251.565\ldots^\circ$ (in range)

Now solve $\cot\theta = -1$:

$\Rightarrow \frac{1}{\tan\theta} = -1 \Rightarrow \tan\theta = -1$

Find first value:

$\theta = \arctan(-1) = -45^\circ$ (not in range)



Add period:

$\theta = -45^\circ + 180^\circ = 135^\circ$ (in range)

$\theta = 135^\circ + 180^\circ = 315^\circ$ (in range)

$\theta = \{71.6^\circ, 135^\circ, 252^\circ, 315^\circ\}$ (3 SF)

28 b)

$4\sec^2\theta - 4\tan\theta - 19 = 0$ for $0 < \theta < 2\pi$

$\Rightarrow 4(\tan^2\theta + 1) - 4\tan\theta - 19 = 0$

$\Rightarrow 4\tan^2\theta - 4\tan\theta - 15 = 0$

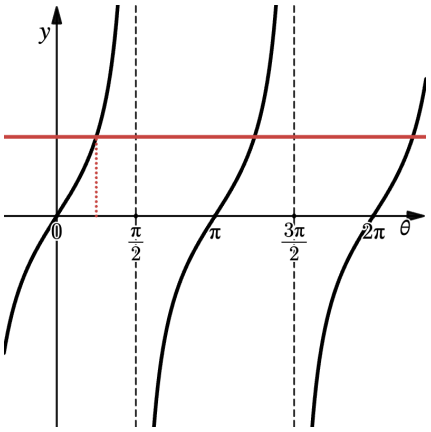
$\Rightarrow (2\tan\theta - 5)(2\tan\theta + 3) = 0$

$\Rightarrow \tan\theta = \frac{5}{2}$ or $\tan\theta = -\frac{3}{2}$

Solve $\tan\theta = \frac{5}{2}$ first:

Find first value:

$\theta = \arctan\frac{5}{2} = 1.1902\ldots$ (in range)



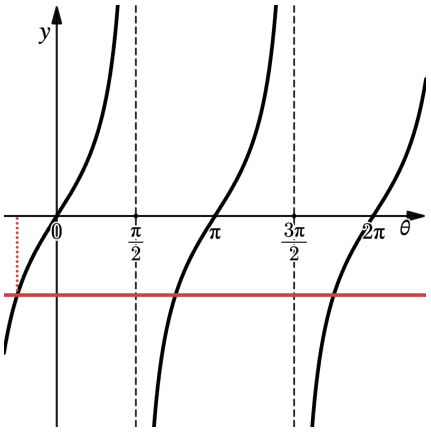
Add period:

$\theta = 1.1902\ldots + \pi = 4.3318\ldots$ (in range)

Now solve $\tan\theta = -\frac{3}{2}$:

Find first value:

$\theta = \arctan\left(-\frac{3}{2}\right) = -0.9827\ldots$ (not in range)



Add period:

$\theta = -0.9827\ldots + \pi = 2.1587\ldots$ (in range)

$\theta = 2.1587\ldots + \pi = 5.3003\ldots$ (in range)

$\theta = \{1.19, 2.16, 4.33, 5.30\}$ (3 SF)

28 c)

$6\cot^2\theta + 23\operatorname{cosec}\theta + 13 = 0$ for $0° < \theta < 360°$

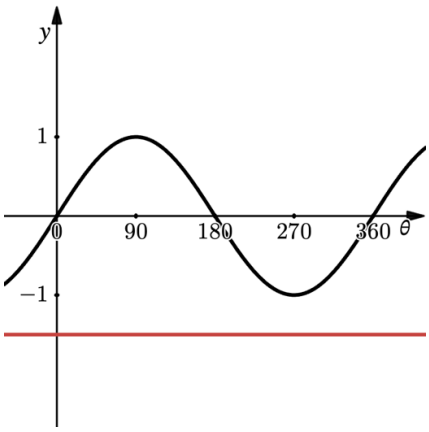
$\Rightarrow 6(\operatorname{cosec}^2\theta - 1) + 23\operatorname{cosec}\theta + 13 = 0$

$\Rightarrow 6\operatorname{cosec}^2\theta + 23\operatorname{cosec}\theta + 7 = 0$

$\Rightarrow (3\operatorname{cosec}\theta + 1)(2\operatorname{cosec}\theta + 7) = 0$

$\Rightarrow \operatorname{cosec}\theta = -\frac{1}{3}$ or $\operatorname{cosec}\theta = -\frac{7}{2}$

As for $\operatorname{cosec}\theta = -\frac{1}{3} \Rightarrow \sin\theta = -3$, there are no solutions:

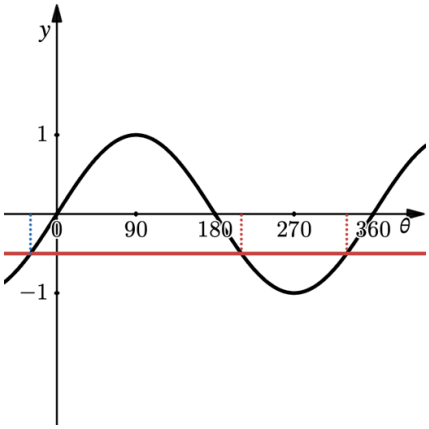


Now solve $\operatorname{cosec}\theta = -\frac{7}{2}$:

$\Rightarrow \frac{1}{\sin\theta} = -\frac{7}{2} \Rightarrow \sin\theta = -\frac{2}{7}$

Find first value:

$\theta = \arcsin\left(-\frac{2}{7}\right) = -16.601\ldots^\circ$ (not in range)



Find first value between 0° and 360°:

$\theta = 180^\circ + 16.601\ldots^\circ = 196.601\ldots^\circ$ (in range)

Find second value between 0° and 360°:

$\theta = 360^\circ - 16.601\ldots^\circ = 343.398\ldots^\circ$ (in range)

$\theta = \{197^\circ, 343^\circ\}$ (3 SF)

28 d)

$18\tan^2\theta + 9\sec\theta - 2 = 0$ for $0 < \theta < 2\pi$

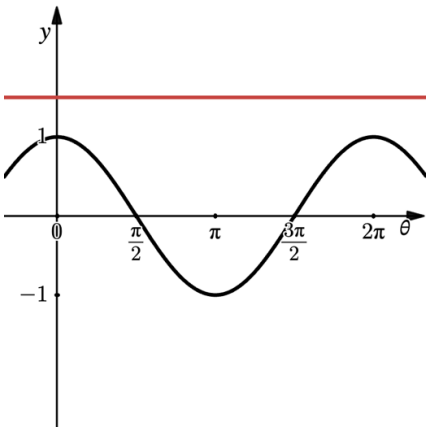
$\Rightarrow 18(\sec^2\theta - 1) + 9\sec\theta - 2 = 0$

$\Rightarrow 18\sec^2\theta + 9\sec\theta - 20 = 0$

$\Rightarrow (6\sec\theta - 5)(3\sec\theta + 4) = 0$

$\Rightarrow \sec\theta = \frac{5}{6}$ or $\sec\theta = -\frac{4}{3}$

As for $\sec\theta = \frac{5}{6} \Rightarrow \cos\theta = \frac{6}{5}$, there are no solutions:

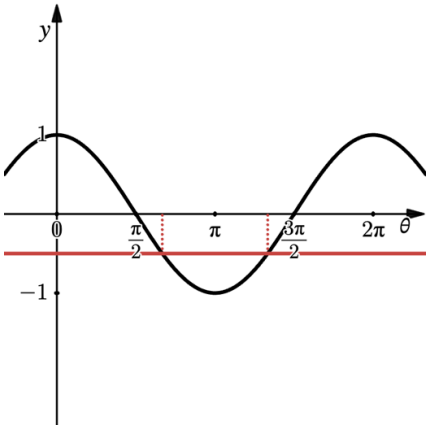


Now solve $\sec\theta = -\frac{4}{3}$:

$\Rightarrow \frac{1}{\cos\theta} = -\frac{4}{3} \Rightarrow \cos\theta = -\frac{3}{4}$

Find first value between 0 and 2π:

$\theta = \arccos\left(-\frac{3}{4}\right) = 2.4188\ldots$ (in range)



Find second value between 0 and 2π:

$\theta = 2\pi - 2.4188\ldots = 3.8643\ldots$ (in range)

$\theta = \{2.42, 3.86\}$ (3 SF)

29)

$3\tan^2 x + 4k\sec x + \frac{4k^2}{3} + 3$

$\equiv 3(\sec^2 x - 1) + 4k\sec x + \frac{4k^2}{3} + 3 = 0$

$\equiv 3\sec^2 x + 4k\sec x + \frac{4k^2}{3}$

$\equiv 3\left(\sec^2 x + \frac{4k}{3}\sec x\right) + \frac{4k^2}{3}$

$\equiv 3\left(\left(\sec x + \frac{2k}{3}\right)^2 - \frac{4k^2}{9}\right) + \frac{4k^2}{3}$

$\equiv 3\left(\sec x + \frac{2k}{3}\right)^2 - \frac{4k^2}{3} + \frac{4k^2}{3}$

$\equiv 3\left(\sec x + \frac{2k}{3}\right)^2$

30 a)

$10\tan^2 x + 17\sec x + 16 = 0$

$\Rightarrow 10(\sec^2 x - 1) + 17\sec x + 16 = 0$

$\Rightarrow 10\sec^2 x + 17\sec x + 6 = 0$

$\Rightarrow (5\sec x + 6)(2\sec x + 1) = 0$

$\Rightarrow \sec x = -\frac{6}{5}$ or $\sec x = -\frac{1}{2}$

$y = \sec x$ only exists for $y \leq -1$ or $y \geq 1$

so $\sec x \neq -\frac{1}{2}$

$\therefore \sec x = -\frac{6}{5}$

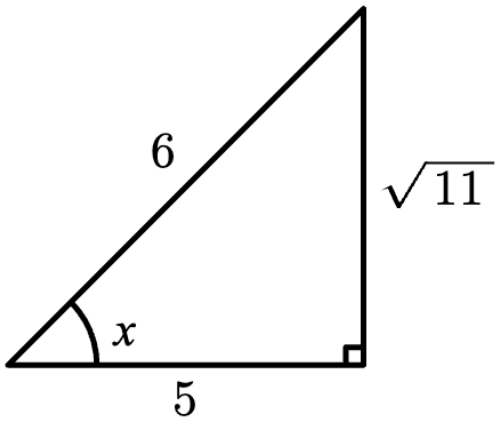
30 b)

$\cos x = \frac{1}{-\frac{6}{5}} = -\frac{5}{6}$

30 c)

x is **OBTUSE**

Ignore $90° < x < 180°$ for the moment and draw a right-angled triangle with this information:



$\sin x = \frac{\sqrt{11}}{6}$ (the result is positive as $\sin x$ is above the x-axis between 90° and 180°)

30 d)

$\tan x = -\frac{\sqrt{11}}{5}$ (the result is negative as $\tan x$ is below the x-axis between 90° and 180°)

30 e)

$\operatorname{cosec} x = \frac{1}{\sin x} = \frac{1}{\frac{\sqrt{11}}{6}} = \frac{6}{\sqrt{11}} = \frac{6\sqrt{11}}{11}$

30 f)

$\cot x = \frac{1}{\tan x} = \frac{1}{-\frac{\sqrt{11}}{5}} = -\frac{5}{\sqrt{11}} = -\frac{5\sqrt{11}}{11}$

31 a)

$16\cot^2 x + 32\operatorname{cosec} x + 31 = 0$

$\Rightarrow 16(\operatorname{cosec}^2 x - 1) + 32\operatorname{cosec} x + 31 = 0$

$\Rightarrow 16\operatorname{cosec}^2 x + 32\operatorname{cosec} x + 15 = 0$

$\Rightarrow (4\operatorname{cosec} x + 5)(4\operatorname{cosec} x + 3) = 0$

$\Rightarrow \operatorname{cosec} x = -\frac{5}{4}$ or $\operatorname{cosec} x = -\frac{3}{4}$

$y = \operatorname{cosec} x$ only exists for $y \leq -1$ or $y \geq 1$

so $\operatorname{cosec} x \neq -\frac{3}{4}$

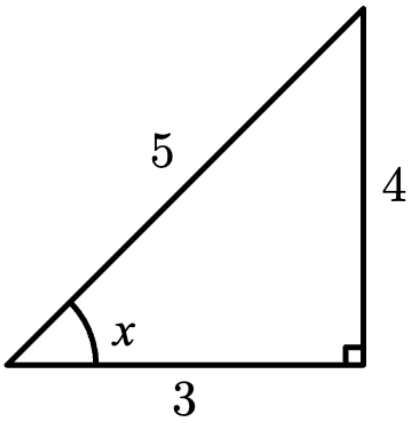
$\therefore \operatorname{cosec} x = -\frac{5}{4}$

31 b)

$\sin x = \frac{1}{-\frac{5}{4}} = -\frac{4}{5}$

31 c)

Ignore $\frac{3\pi}{2} < x < 2\pi$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\cos x = \frac{3}{5}$ (the result is positive as $\cos x$ is above the x-axis between $\frac{3\pi}{2}$ and 2π)

31 d)

$\tan x = -\frac{4}{3}$ (the result is negative as $\tan x$ is below the x-axis between $\frac{3\pi}{2}$ and 2π)

31 e)

$\sec x = \frac{1}{\cos x} = \frac{1}{\frac{3}{5}} = \frac{5}{3}$

31 f)

$\cot x = \frac{1}{\tan x} = \frac{1}{-\frac{4}{3}} = -\frac{3}{4}$

32)

$12\cot^2\theta - 5\operatorname{cosec}\theta + 10 = 0$

$\Rightarrow 12(\operatorname{cosec}^2\theta - 1) - 5\operatorname{cosec}\theta + 10 = 0$

$\Rightarrow 12\operatorname{cosec}^2\theta - 5\operatorname{cosec}\theta - 2 = 0$

$\Rightarrow (3\operatorname{cosec}\theta - 2)(4\operatorname{cosec}\theta + 1) = 0$

$\Rightarrow \operatorname{cosec}\theta = \frac{2}{3}$ or $\operatorname{cosec}\theta = -\frac{1}{4}$

$y = \operatorname{cosec} x$ only exists for $y \leq -1$ or $y \geq 1$

So $12\cot^2\theta - 5\operatorname{cosec}\theta + 10 = 0$ has no real solutions

33 a)

$\mathrm{p}\left(-\frac{2}{3}\right) = 12\left(-\frac{2}{3}\right)^3 + 8\left(-\frac{2}{3}\right)^2 - 3\left(-\frac{2}{3}\right) - 2$

$= -\frac{32}{9} + \frac{32}{9} + 2 - 2 = 0 \therefore (3x+2)$ is a factor of $\mathrm{p}(x)$ by the Factor Theorem

33 b)

	$4x^2$	$0x$	-1
$3x$	$12x^3$	$0x^2$	$-3x$
2	$8x^2$	$0x$	-2

$\therefore \mathrm{p}(x) = (3x+2)(4x^2-1)$

$\Rightarrow \mathrm{p}(x) = (3x+2)(2x-1)(2x+1)$

33 c)

$12\,\mathrm{cosec}^3\,\theta + 8\cot^2\theta - 3\,\mathrm{cosec}\,\theta + 6 = 0$

$\Rightarrow 12\,\mathrm{cosec}^3\,\theta + 8(\mathrm{cosec}^2\,\theta - 1) - 3\,\mathrm{cosec}\,\theta + 6 = 0$

$\Rightarrow 12\,\mathrm{cosec}^3\,\theta + 8\,\mathrm{cosec}^2\,\theta - 3\,\mathrm{cosec}\,\theta - 2 = 0$

$\Rightarrow (3\,\mathrm{cosec}\,\theta + 2)(2\,\mathrm{cosec}\,\theta - 1)(2\,\mathrm{cosec}\,\theta + 1) = 0$

$\Rightarrow \mathrm{cosec}\,\theta = -\frac{2}{3}$ or $\mathrm{cosec}\,\theta = \frac{1}{2}$ or $\mathrm{cosec}\,\theta = -\frac{1}{2}$

$y = \mathrm{cosec}\,x$ only exists for $y \leq -1$ or $y \geq 1$

So $12\,\mathrm{cosec}^3\,\theta + 8\cot^2\theta - 3\,\mathrm{cosec}\,\theta + 6 = 0$ has no real solutions

34 a)

LHS: $\cot x + \cos x \equiv \frac{\cos x}{\sin x} + \frac{\cos x}{1} \equiv \frac{\cos x + \cos x \sin x}{\sin x}$

$\equiv \frac{\cos x(1+\sin x)}{\sin x} \equiv \frac{\cos x}{\sin x} \times (1+\sin x)$

$\equiv \cot x(1+\sin x)$:RHS

34 b)

LHS: $\cot x + \sec x \equiv \frac{\cos x}{\sin x} + \frac{1}{\cos x} \equiv \frac{\cos^2 x + \sin x}{\sin x \cos x}$

$\equiv \frac{1}{\sin x} \times \frac{1}{\cos x} \times (\sin x + \cos^2 x)$

$\equiv \mathrm{cosec}\,x \sec x(\sin x + \cos^2 x)$:RHS

34 c)

LHS: $\cot x + \sin x \equiv \frac{\cos x}{\sin x} + \frac{\sin x}{1} \equiv \frac{\cos x + \sin^2 x}{\sin x}$

$\equiv \frac{1}{\sin x} \times (\sin^2 x + \cos x)$

$\equiv \mathrm{cosec}\,x(\sin^2 x + \cos x)$:RHS

34 d)

LHS: $\cot x + \tan x \equiv \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} \equiv \frac{\cos^2 x + \sin^2 x}{\sin x \cos x}$

$\equiv \frac{1}{\sin x \cos x} \equiv \frac{1}{\sin x} \times \frac{1}{\cos x} \equiv \mathrm{cosec}\,x \sec x$:RHS

35 a)

LHS: $\sec x - \cos x \equiv \frac{1}{\cos x} - \frac{\cos x}{1} \equiv \frac{1-\cos^2 x}{\cos x} \equiv \frac{\sin^2 x}{\cos x}$

$\equiv \sin x \times \frac{\sin x}{\cos x} \equiv \sin x \tan x$:RHS

35 b)

LHS: $\operatorname{cosec} x - \sin x \equiv \frac{1}{\sin x} - \frac{\sin x}{1} \equiv \frac{1-\sin^2 x}{\sin x} \equiv \frac{\cos^2 x}{\sin x}$

$\equiv \cos x \times \frac{\cos x}{\sin x} \equiv \cos x \cot x$:RHS

35 c)

LHS: $\tan x - \cot x \equiv \frac{\sin x}{\cos x} - \frac{\cos x}{\sin x} \equiv \frac{\sin^2 x - \cos^2 x}{\sin x \cos x}$

$\equiv \frac{\sin^2 x - (1-\sin^2 x)}{\sin x \cos x} \equiv \frac{2\sin^2 x - 1}{\sin x \cos x}$

$\equiv \frac{1}{\sin x} \times \frac{1}{\cos x} \times (2\sin^2 x - 1)$

$\equiv \operatorname{cosec} x \sec x (2\sin^2 x - 1)$:RHS

35 d)

LHS: $\tan x - \operatorname{cosec} x \equiv \frac{\sin x}{\cos x} - \frac{1}{\sin x} \equiv \frac{\sin^2 x - \cos x}{\sin x \cos x}$

$\equiv \frac{1}{\sin x} \times \frac{\sin^2 x - \cos x}{\cos x} \equiv \frac{1}{\sin x}\left(\frac{\sin^2 x}{\cos x} - \frac{\cos x}{\cos x}\right)$

$\equiv \operatorname{cosec} x \left(\sin x \times \frac{\sin x}{\cos x} - 1\right)$

$\equiv \operatorname{cosec} x (\sin x \tan x - 1)$:RHS

36 a)

LHS: $\sin^2 x - \cot^2 x \equiv \frac{\sin^2 x}{1} - \frac{\cos^2 x}{\sin^2 x} \equiv \frac{\sin^4 x - \cos^2 x}{\sin^2 x}$

$\equiv \frac{1}{\sin^2 x} \times (\sin^4 x - \cos^2 x)$

$\equiv \operatorname{cosec}^2 x (\sin^4 x - \cos^2 x)$:RHS

36 b)

LHS: $\cos^2 x - \cot^2 x \equiv \frac{\cos^2 x}{1} - \frac{\cos^2 x}{\sin^2 x}$

$\equiv \frac{\sin^2 x \cos^2 x - \cos^2 x}{\sin^2 x} \equiv \frac{\cos^2 x (\sin^2 x - 1)}{\sin^2 x}$

$\equiv \frac{\cos^2 x}{\sin^2 x} \times (\sin^2 x - 1) \equiv -\frac{\cos^2 x}{\sin^2 x} \times (1 - \sin^2 x)$

$\equiv -\cot^2 x \cos^2 x$:RHS

36 c)

LHS: $\sec^2 x - \operatorname{cosec}^2 x \equiv \frac{1}{\cos^2 x} - \frac{1}{\sin^2 x}$

$\equiv \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} \equiv \frac{1}{\sin^2 x} \times \frac{1}{\cos^2 x} \times (\sin^2 x - \cos^2 x)$

$\equiv \frac{1}{\sin^2 x} \times \frac{1}{\cos^2 x} \times (\sin^2 x - \cos^2 x)$

$\equiv \operatorname{cosec}^2 x \sec^2 x (\sin^2 x - (1 - \sin^2 x))$

$\equiv \operatorname{cosec}^2 x \sec^2 x (2\sin^2 x - 1)$:RHS

36 d)

LHS: $\tan^2 x - \cot^2 x \equiv \frac{\sin^2 x}{\cos^2 x} - \frac{\cos^2 x}{\sin^2 x} \equiv \frac{\sin^4 x - \cos^4 x}{\sin^2 x \cos^2 x}$

$\equiv \frac{1}{\sin^2 x} \times \frac{1}{\cos^2 x} \times (\sin^4 x - \cos^4 x)$

$\equiv \operatorname{cosec}^2 x \sec^2 x (\sin^2 x + \cos^2 x)(\sin^2 x - \cos^2 x)$

$\equiv \operatorname{cosec}^2 x \sec^2 x (\sin^2 x - \cos^2 x) \equiv$

$\equiv \operatorname{cosec}^2 x \sec^2 x (\sin^2 x - (1 - \sin^2 x))$

$\equiv \operatorname{cosec}^2 x \sec^2 x (2\sin^2 x - 1)$:RHS

37 a)

LHS: $(\sec x + \cot x)^2$

$\equiv \sec^2 x + 2 \sec x \cot x + \cot^2 x$

$\equiv (\tan^2 x + 1) + 2 \times \frac{1}{\cos x} \times \frac{\cos x}{\sin x} + (\operatorname{cosec}^2 x - 1)$

$\equiv \tan^2 x + \operatorname{cosec}^2 x + \frac{2}{\sin x}$

$\equiv \tan^2 x + 2 \operatorname{cosec} x + \operatorname{cosec}^2 x$:RHS

37 b)

LHS: $(\sec x + \operatorname{cosec} x)^2$

$\equiv \sec^2 x + 2 \sec x \operatorname{cosec} x + \operatorname{cosec}^2 x$

$\equiv (\tan^2 x + 1) + 2 \times \frac{1}{\cos x} \times \frac{1}{\sin x} + (1 + \cot^2 x)$

$\equiv \tan^2 x + \cot^2 x + 2\left(\frac{1}{\cos x \sin x} + 1\right)$

$\equiv \tan^2 x + \cot^2 x + 2\left(\frac{\sin^2 x + \cos^2 x}{\cos x \sin x} + 1\right)$

$\equiv \tan^2 x + \cot^2 x + 2\left(\frac{\sin^2 x}{\cos x \sin x} + \frac{\cos^2 x}{\cos x \sin x} + 1\right)$

$\equiv \tan^2 x + \cot^2 x + 2\left(\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} + 1\right)$

$\equiv \tan^2 x + \cot^2 x + 2(\tan x + \cot x + 1)$:RHS

37 c)

LHS: $(\operatorname{cosec} x + \cot x)^2$

$\equiv \operatorname{cosec}^2 x + 2 \operatorname{cosec} x \cot x + \cot^2 x$

$\equiv (\cot^2 x + 1) + 2 \times \frac{1}{\sin x} \times \frac{\cos x}{\sin x} + \cot^2 x$

$\equiv 2\cot^2 x + 1 + \frac{2\cos x}{\sin^2 x}$

$\equiv 2\cot^2 x + 1 + 2\cos x \times \frac{1}{\sin^2 x}$

$\equiv 2\cot^2 x + 1 + 2\cos x \times \left(\frac{\sin^2 x + \cos^2 x}{\sin^2 x}\right)$

$\equiv 2\cot^2 x + 1 + 2\cos x \times \left(\frac{\sin^2 x}{\sin^2 x} + \frac{\cos^2 x}{\sin^2 x}\right)$

$\equiv 2\cot^2 x + 1 + 2\cos x(\cot^2 x + 1)$:RHS

37 d)

LHS: $(\tan x + \sec x)^2$

$\equiv \tan^2 x + 2 \tan x \sec x + \sec^2 x$

$\equiv (\sec^2 x - 1) + 2 \times \frac{\sin x}{\cos x} \times \frac{1}{\cos x} + \sec^2 x$

$\equiv 2\sec^2 x - 1 + \frac{2\sin x}{\cos^2 x}$

$\equiv 2\sec^2 x - 1 + 2\sin x \times \frac{1}{\cos^2 x}$

$\equiv 2\sec^2 x - 1 + 2\sin x \sec^2 x$

$\equiv 2\sec^2 x(\sin x + 1) - 1$:RHS

37 e)

LHS: $(\tan x + \sec x)^2 \equiv \left(\frac{\sin x}{\cos x} + \frac{1}{\cos x}\right)^2 \equiv \left(\frac{\sin x+1}{\cos x}\right)^2$

$\equiv \frac{(\sin x+1)^2}{\cos^2 x} \equiv \frac{(\sin x+1)^2}{1-\sin^2 x} \equiv \frac{(1+\sin x)^2}{(1-\sin x)(1+\sin x)}$

$\equiv \frac{1+\sin x}{1-\sin x}$:RHS

38 a)

LHS: $\frac{\sin\theta}{1-\cos\theta} - \operatorname{cosec}\theta \equiv \frac{\sin\theta}{1-\cos\theta} \times \frac{1+\cos\theta}{1+\cos\theta} - \frac{1}{\sin\theta}$

$\equiv \frac{\sin\theta(1+\cos\theta)}{1-\cos^2\theta} - \frac{1}{\sin\theta} \equiv \frac{\sin\theta(1+\cos\theta)}{\sin^2\theta} - \frac{1}{\sin\theta}$

$\equiv \frac{1+\cos\theta}{\sin\theta} - \frac{1}{\sin\theta} \equiv \frac{1+\cos\theta-1}{\sin\theta} \equiv \frac{\cos\theta}{\sin\theta} \equiv \cot\theta$:RHS

38 b)

$\frac{\sin\theta}{1-\cos\theta} = 10 + \cot\theta$ for $0 < \theta < 2\pi$

$\Rightarrow \frac{\sin\theta}{1-\cos\theta} - \operatorname{cosec}\theta = 10 + \cot\theta - \operatorname{cosec}\theta$

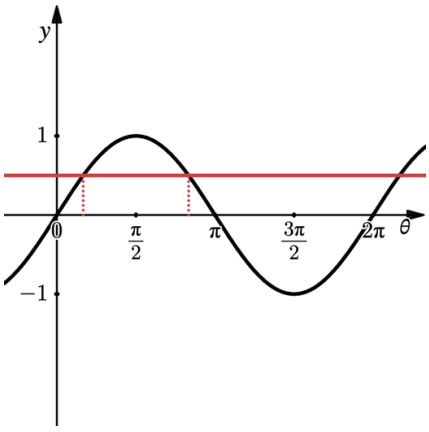
$\Rightarrow \cot\theta = 10 + \cot\theta - \operatorname{cosec}\theta$

$\Rightarrow 0 = 10 - \operatorname{cosec}\theta \Rightarrow \operatorname{cosec}\theta = 10$

$\Rightarrow \frac{1}{\sin\theta} = 10 \Rightarrow \sin\theta = \frac{1}{10}$

Find first value between 0 and 2π:

$\theta = \arcsin\frac{1}{10} = 0.1001\ldots$ (in range)



Find second value between 0 and 2π:

$\theta = \pi - 0.1001\ldots = 3.0414\ldots$ (in range)

$\theta = \{0.100, 3.04\}$ (3 SF)

39)

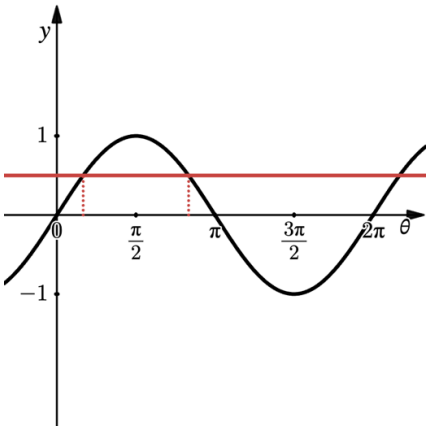
$\operatorname{cosec}\theta = 10\cos\theta$ for $0 < \theta < 2\pi$

$\Rightarrow \frac{1}{\sin\theta} = 10\cos\theta \Rightarrow 1 = 10\sin\theta\cos\theta$

$\Rightarrow 1 = 5(2\sin\theta\cos\theta) \Rightarrow 1 = 5\sin 2\theta \Rightarrow \sin 2\theta = \frac{1}{5}$

Find first value between 0 and 2π:

$2\theta = \arcsin\frac{1}{5} = 0.2013\ldots$



Find second value between 0 and 2π:

$2\theta = \pi - 0.2013\ldots = 2.9402\ldots$

Apply transformation:

$2\theta = 0.2013\ldots \Rightarrow \theta = 0.1006\ldots$ (in range)

$2\theta = 2.9402\ldots \Rightarrow \theta = 1.4701\ldots$ (in range)

Add period:

$\theta = 0.1006\ldots + \pi = 3.2422\ldots$ (in range)

$\theta = 1.4701\ldots + \pi = 4.6117\ldots$ (in range)

$\theta = \{0.101, 1.47, 3.24, 4.61\}$ (3 SF)